

ADVANCED MATHEMATICAL MODELS IN FINANCE

Final Project Part B

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1 Question 1

The objective of this analysis is to compare a traditional indemnity insurance contract with a parametric insurance contract using the mean–variance (MV) criterion.

Flood losses are modelled through the aggregate annual loss

$$S = \sum_{i=1}^N Y_i \quad (1)$$

where N denotes the number of flood events occurring during the year and Y_i represent monetary damage caused by the i -th flood. The claim count N follows a Poisson distribution. We assume that N is independent of the *i.i.d.* severities $\{Y_i\}_{i \geq 1}$ where each Y_i follows a censored exponential distribution with density reported in Eq 4.

The comparison is particularly relevant because the two insurance products transfer risk in fundamentally different ways. A traditional indemnity contract compensates realized losses above a deductible d

$$B^{(d)} = \sum_{i=1}^N (Y_i - d)^+ \quad (2)$$

thus protecting the right tail of the loss distribution. In contrast, the parametric contract pays a fixed amount k whenever a flood related event occurs

$$B^{(p)} = kN \quad (3)$$

independently of the realized loss severity. Therefore, the parametric contract provides first-dollar protection but introduces basis risk, since the payout k may differ from the actual loss Y_i .

Unlike indemnity insurance, the parametric contract does not require claims assessment or loss verification. This operational simplicity can be represented by lower fixed costs $\gamma_p < \gamma_d$ and potentially lower loading factors $\theta_p < \theta_d$, reflecting lower capital requirements and settlement expenses. The purpose of this report is to determine under which conditions the welfare gain from lower costs outweighs the basis risk introduced by the parametric design.

1.1 Comparison of Insurance Approaches

To evaluate the threshold where a decision-maker becomes indifferent between options, we analyze two distinct structural frameworks for the parametric insurance contract:

- Approach (a): Optimally Chosen Parametric Insurance (k^*):
The parametric payout factor or coverage limit k is chosen to maximize the decision-maker's expected utility, or minimize the variance of the residual risk (basis risk).
- Approach (b): Cost-Equivalent Parametric Insurance:
Instead of optimizing explicitly for risk reduction, the coverage parameter k is fixed in advance. It is calibrated so that the premium of the parametric contract matches the premium of the corresponding optimal classical indemnity contract.

1.2 Theoretical Foundations

The problem set the following values: a shed value $L = \text{€}1\,000\,000$, an initial wealth $\omega_0 = 0.2 \cdot L = \text{€}200\,000$, and a risk-aversion parameter $\beta = \frac{1}{\omega_0} = \frac{1}{200\,000}$.

The number of flood events N follows a Poisson distribution with intensity $\lambda = \frac{1}{40}$, while individual claim severities Y_i follow a censored exponential distribution with upper bound L , whose density is

$$f_{Y_i}(y) = \begin{cases} \nu e^{-\nu y} & y \in [0, L) \\ e^{-\nu L} \delta(y - L) & y = L \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

where $\nu = \frac{1}{700\,000}$ is the severity parameter of the censored exponential distribution.

Quantity	Interpretation	Units
k, d, L, ω_0	monetary values	€
P_d, P_p, MV	premiums / objective	€
λ	claim frequency	year ⁻¹
ν	severity scale	€ ⁻¹
θ_d, θ_p	loading factors	dimensionless
β	risk aversion	€ ⁻¹

Table 1: Model parameters and units.

1.2.1 Parametric Insurance Contract Premium

The premium of the parametric insurance contract is assumed to scale linearly with the selected coverage factor k . Following the pricing framework of [1], the fixed-cost component γ_p is included within the expected-value premium principle and is therefore subject to the loading factor $(1 + \theta_p)$. This reflects the assumption that administrative and operational costs require risk capital and consequently contribute to the insurer's risk-adjusted premium. The parametric premium is therefore given by

$$P_p = (1 + \theta_p) (\lambda k + \gamma_p), \quad (5)$$

where λk denotes the expected payout and θ_p is the premium loading factor. For simplicity, we assume that the fixed-cost component is negligible, i.e., $\gamma_p = 0$. Hence, the premium reduces to

$$P_p = (1 + \theta_p) \lambda k. \quad (6)$$

1.2.2 Parametric Mean-Variance (MV) utility function

In parametric insurance, the contract does not reimburse the realized loss; instead, it pays a fixed amount k for each event that exceeds an objective threshold. This structure eliminates the need for loss assessment, reduces the insurer's operational costs, and enables rapid post-event payouts. The contract introduces basis risk, the divergence between the fixed payment k and the actual loss Y_i , but remains scalable and accessible even under stringent budget constraints, precisely when traditional indemnity insurance becomes economically irrelevant due to high deductibles. To evaluate the contract performance, we establish the insured's mean-variance objective function

$$MV = \mathbb{E}[W] - \beta \cdot \text{Var}(W) \quad (7)$$

The terminal wealth of the insured under the parametric contract is:

$$W = \omega_0 - P_p - S + kN \quad (8)$$

Considering our setting where $P_p = (1 + \theta_p) \lambda k$, and $S = \sum_{i=1}^N Y_i$, the expected mean-variance of the parametric cover is:

$$MV_p = \omega_0 - P_p - \lambda \mathbb{E}[Y_i] + \lambda k - \beta \lambda (\text{Var}(Y_i) + (E[Y_i] - k)^2) \quad (9)$$

As presented in [1] in Equation (3.8).

Considering the relationship $\text{Var}(Y_i) = \mathbb{E}(Y_i^2) - \mathbb{E}(Y_i)^2$ for a given random variable, this can be written as:

$$MV_p = \omega_0 - P_p - \lambda \mathbb{E}[Y_i] + \lambda k - \beta \lambda (\mathbb{E}[Y_i^2] - 2k \mathbb{E}[Y_i] + k^2) \quad (10)$$

Taking the first-derivative with respect to k and imposing it equal to zero:

$$\frac{\partial MV_p}{\partial k} = \lambda - (1 + \theta_p) \lambda + 2\beta \lambda (\mathbb{E}[Y_i] - k) = 0 \quad (11)$$

Dividing through by $\lambda > 0$:

$$-\theta_p + 2\beta(\mathbb{E}[Y_i] - k) = 0 \quad (12)$$

The second-order condition $\frac{\partial^2 MV_p}{\partial k^2} = -2\beta \lambda < 0$ confirms this is the unique maximiser.

$$k^* = \mathbb{E}[Y_i] - \frac{\theta_p}{2\beta} \quad (13)$$

$$\text{with } \theta_p = 0.30 \Rightarrow k^* = 532\,244,14 - \frac{0.3}{\frac{2}{200000}} = \text{€}502\,244.14$$

$$\mathbb{E}[Y] = \int_0^L y \cdot v \cdot e^{-vy} dy + L \cdot e^{-vL}$$

Using integration by parts: $\int u \cdot v' = u \cdot v - \int u' \cdot v$ then we have

$$= -L \cdot e^{-vL} + \frac{1}{v} (1 - e^{-vL})$$

$$\mathbb{E}[Y] = -L \cdot e^{-vL} + \frac{1}{v} (1 - e^{-vL}) + L \cdot e^{-vL}$$

$$\mathbb{E}[Y] = \frac{1}{v} (1 - e^{-vL}) = 532\,244,14$$

The second moment of the censored exponential severity is:

$$\mathbb{E}[Y_i^2] = \int_0^L y^2 \nu e^{-\nu y} dy + L^2 e^{-\nu L} \quad (14)$$

Applying integration by parts twice to the integral:

$$\begin{aligned} \int_0^L y^2 \nu e^{-\nu y} dy &= [-y^2 e^{-\nu y}]_0^L + \int_0^L 2y e^{-\nu y} dy \\ &= -L^2 e^{-\nu L} + \frac{2}{\nu} [-y e^{-\nu y}]_0^L + \frac{2}{\nu} \int_0^L e^{-\nu y} dy \\ &= -L^2 e^{-\nu L} - \frac{2L}{\nu} e^{-\nu L} + \frac{2}{\nu^2} (1 - e^{-\nu L}) \end{aligned} \quad (15)$$

Adding back the atom $L^2 e^{-\nu L}$:

$$\mathbb{E}[Y_i^2] = \frac{2}{\nu^2} (1 - e^{-\nu L}) - \frac{2L}{\nu} e^{-\nu L} \quad (16)$$

Substituting ν and L in the formulas:

$$\mathbb{E}[Y_i^2] = 2 \cdot (700,000)^2 (1 - e^{-10/7}) - 2 \cdot 700,000 \cdot e^{-10/7} \approx 409,630,533,268.57$$

With $\theta_p = 0.3$

$$\begin{aligned} MV_p &= 200,000 - 16,322.93 - \frac{532,244.14}{40} + \frac{502,244.14}{40} \\ &\quad - \frac{1}{8,000,000} [409,630,533,268.57 + (502,244.14)^2 \\ &\quad - 2(502,244.14)(532,244.14)] \\ &= 182,927.07 - 15,905.82 = \text{€}167,021.25 \end{aligned}$$

With $\theta_p = 0.2$

$$\begin{aligned} MV_p &= 200,000 - 15,367.32 - \frac{532,244.14}{40} + \frac{512,244.14}{40} \\ &\quad - \frac{1}{8,000,000} [409,630,533,268.57 + (512,244.14)^2 \\ &\quad - 2(512,244.14)(532,244.14)] \\ &= 183,827.07 - 15,538.32 = \text{€}168,288.76 \end{aligned}$$

1.2.3 Approach (a): Parametric Insurance k^* chosen optimally

$$\text{with } \theta_p = 0.3 \Rightarrow P_p = (1 + 0.3) \cdot \frac{1}{40} \cdot 502\,244.14 + 0 = \text{€}16\,322.93$$

$$\text{with } \theta_p = 0.2 \Rightarrow P_p = (1 + 0.2) \cdot \frac{1}{40} \cdot 512\,244.14 + 0 = \text{€}15\,367.32$$

1.2.4 Classic Insurance Mean-Variance (MV) utility function

$$W = \omega_0 - P_d - \sum_{i=1}^N \min(Y_i, d) \quad (17)$$

$$\text{Var}(W) = \text{Var}\left(\sum_{i=1}^N \min(Y_i, d)\right) = \lambda \cdot E[\min(Y_i, d)^2]$$

The insured retains losses below the deductible d , so the residual loss per claim is $\min(Y_i, d)$, while the insurer covers $(Y_i - d)^+$. We can decompose:

$$Y_i = \min(Y_i, d) + (Y_i - d)^+ \Rightarrow \mathbb{E}[Y_i] = \mathbb{E}[\min(Y_i, d)] + \mathbb{E}[(Y_i - d)^+]$$

so that:

$$\lambda \mathbb{E}[\min(Y_i, d)] = \lambda \mathbb{E}[Y_i] - \lambda \mathbb{E}[(Y_i - d)^+]$$

Since the terminal wealth is

$$W = \omega_0 - P_d - \sum_{i=1}^N \min(Y_i, d) \quad (18)$$

we obtain

$$\mathbb{E}[W] = \omega_0 - P_d - \lambda \mathbb{E}[\min(Y_i, d)] = \omega_0 - P_d - \lambda \mathbb{E}[Y_i] + \lambda \mathbb{E}[(Y_i - d)^+] \quad (19)$$

$$\text{Var}(W) = \lambda \mathbb{E}[\min(Y_i, d)^2] =: \lambda G(d) \quad (20)$$

$$MV_d(\gamma_d) = \omega_0 - P_d(\gamma_d) - \lambda \mathbb{E}[Y_i] + \lambda \mathbb{E}[(Y_i - d)^+] - \beta \lambda G(d) \quad (21)$$

where $G(d)$ follows the following distribution as obtained in [1]:

$$G(d) = \frac{2}{v^2} [1 - e^{-vd}(1 + vd)] \quad (22)$$

Taking the first-order derivative with respect to d :

$$\frac{\partial MV_d}{\partial d} = \lambda e^{-\nu d} (\theta_d - 2\beta d) = 0$$

Given $\lambda > 0$ and $e^{-\nu d} > 0$, the unique interior maximiser is:

$$d^* = \frac{\theta_d}{2\beta} \quad \left. \frac{\partial^2 MV_d}{\partial d^2} \right|_{d^*} = -2\beta \lambda e^{-\nu d^*} < 0 \quad (23)$$

The result holds for any censored exponential severity and does not depend on the specific value of ν . The effective deductible is then:

$$d^* = \min\left(\frac{\theta_d}{2\beta}, L\right) \quad (24)$$

$$\text{with } \theta_d = 0.3 \Rightarrow d^* = \min\left(\frac{0.3}{\frac{2}{200000}}, 1\,000\,000\right) = \text{€}30\,000$$

The classical utility curve MV_d is strictly invariant to changes in the parametric risk loading factor θ_p , as the structural mechanics of the traditional indemnity contract depend exclusively on the classical premium schedule $P_d(\gamma_d)$ and the residual asset exposure within the deductible bracket. We can now evaluate MV_d

$$\begin{aligned} MV_d(d^*, \gamma_d) &= \omega_0 - P_d(\gamma_d) - \lambda \mathbb{E}[Y_i] + \lambda \mathbb{E}[(Y_i - d)^+] - \beta \lambda G(d) \\ &= 182,812.97 - 1.3\gamma_d \end{aligned}$$

1.2.5 Premium of the classical indemnity insurance Contract:

The classical policy offers full risk coverage above a chosen deductible d , but its premium P_d is burdened by an insurer's loading factor θ_d and fixed administrative/loss-verification costs γ_d :

$$P_d(\gamma_d) = (1 + \theta_d) \cdot (\lambda \mathbb{E}[(Y - d)^+] + \gamma_d) \quad (25)$$

$$P_d(d^*, \gamma_d) = (1 + 0.3) \cdot \left[\frac{1}{40}(502,878.05) + \gamma_d \right] = 16,343.54 + 1.3\gamma_d$$

1.2.6 Approach (b): Budget-Matched Retention Level (k_m):

Scenario where the parametric insurance premium is constrained by the exact same budget as the classical alternative ($P_p = P_d(\gamma_d)$).

$$k(d, \gamma_d) = \min \left\{ \frac{P_d/(1 + \theta_p) - \gamma_p}{\lambda}, L \right\} \quad (26)$$

so that the parametric premium matches the indemnity premium whenever the implied k lies below L . In the case with $\theta_p = 0.3$

$$k_m(\gamma_d) = \frac{16,343.54 + 1.3\gamma_d}{(1 + 0.3) \cdot 0.025}$$

The budget-matched coverage level must be capped at the maximum possible catastrophic loss threshold L :

$$k_m^{\text{constrained}}(\gamma_d) = \min \{502,878.15 + 40\gamma_d, L\} \quad (27)$$

Instead with $\theta_p = 0.2$

$$\begin{aligned} k_m(\gamma_d) &= \frac{16,343.54 + 1.3\gamma_d}{(1 + 0.2) \cdot 0.025} = 544,784.67 + 43.33\gamma_d \\ k_m^{\text{constrained}}(\gamma_d) &= \min \{544,784.67 + 43.33\gamma_d, L\} \end{aligned} \quad (28)$$

1.2.7 Budget-Constrained Utility (MV^{budget})

$$MV_p^{\text{(budget)}}(\gamma_d) = \omega_0 - P_p(k(\gamma_d)) - \lambda \mathbb{E}[Y_i] + \lambda k(\gamma_d) - \beta \lambda (\mathbb{E}[Y_i^2] + k(\gamma_d)^2 - 2k(\gamma_d)\mathbb{E}[Y_i])$$

With $\theta_p = 0.3$

$$MV_p^{\text{(budget } P_d)}} = \begin{cases} 167,021.19 - 0.00634\gamma_d - 0.0002\gamma_d^2 & \text{for } \gamma_d < 12,428 \\ 136,051.15 & \text{for } \gamma_d \geq 12,428 \end{cases}$$

With $\theta_p = 0.2$

$$MV_p^{\text{(budget } P_d)}} = \begin{cases} 168,156.99 - 0.3525\gamma_d - 0.00023472\gamma_d^2 & \text{for } \gamma_d < 10,505 \\ 138,551.15 & \text{for } \gamma_d \geq 10,505 \end{cases}$$

1.2.8 The Indifference Threshold (γ_{indif}): Approach (a)

Represents the exact break-even point where the maximum utility of the classical contract equals the utility of the globally optimized parametric contract ($MV_d = MV_p$). Isolating γ_d yields the closed-form formula:

$$MV_d(\gamma_d) = MV_p \quad (29)$$

$$\text{With } \theta_p = 0.3 \Rightarrow \gamma_{\text{indif}} = \frac{182,812.97 - 167,021.25}{1.3} \approx \text{€}12,147$$

$$\text{With } \theta_p = 0.2 \Rightarrow \gamma_{\text{indif}} = \frac{182,812.97 - 168,288.76}{1.3} \approx \text{€}11,172$$

1.2.9 The Indifference Threshold (γ_{indif}): Approach (b)

$$MV_d(\gamma_d) = MV_p^{\text{(budget } P_d)} \quad (30)$$

With $\theta_p = 0.3$

$$182,812.97 - 1.3\gamma_d = 136,051.15$$

$$\gamma_d \approx \text{€}35,971$$

With $\theta_p = 0.2$

$$182,812.97 - 1.3\gamma_d = 138,551.15$$

$$\gamma_d \approx \text{€}34,048$$

1.3 Implementation in Python

Step 1: Initialization

Set the problem parameters:

- $L = \text{€}1,000,000$: Value of the plant, the maximum possible loss
- $\omega_0 = 0.2 \cdot L = \text{€}200,000$: Initial wealth
- $\lambda = \frac{1}{40} = 0.025$: Loss frequency (statistically, one loss every 40 years)
- $\nu = \frac{1}{700,000}$: The severity parameter for the exponential loss distribution
- $\gamma_p = \text{€}0.0$: Fixed transaction cost for the parametric insurance policy
- $\beta = \frac{1}{\omega_0}$: Coefficient of absolute risk aversion

Step 2: Pre-computing Severity Moments

To simplify the formulas, we can pre-calculate EYI and EYI2, which represent the expected loss $E[Y_i]$ and the second moment $E[Y_i^2]$ respectively.

Step 3: Calculating Optimal Contract Parameters

In this step, we determine the optimal contract parameters, specifically the optimal deductible d^* for the indemnity contract and the optimal coverage amount k^* for the parametric contract.

Step 4: Grid Evaluation over γ_d

To evaluate the impact of the indemnity fixed cost (γ_d), the script generates a linear grid of 6,000 points ranging from $\text{€}0$ to $\text{€}80,000$. For each point, it computes:

1. **Indemnity Metrics:** Premium P_d and Utility MV_d
2. **Parametric Metrics:** Premium P_p and Utility MV_p
3. **Budget-Matched Parametric Contract (MV_p^{budget}):** Where the parametric coverage k is dynamically scaled to match the exact premium budget of the indemnity contract ($P_p = P_d$).

Step 5: Threshold Calculation

The script utilizes a linear interpolation function (`find_crossing`) to find the exact indifference thresholds. Specifically, it calculates

$$\gamma_{\text{indif}} \quad \text{where} \quad MV_d = MV_p$$

and the budget-constrained threshold

$$\gamma_{\text{indif}} \quad \text{where} \quad MV_d = MV_p^{\text{budget}}$$

Step 6: Representation

The final step of the implementation utilizes a dual-axis visualization framework to plot the key financial metrics simultaneously. The left y-axis displays the premium trajectories (P_d, P_p), while the right y-axis captures the corresponding utility curves ($MV_d, MV_p, MV_p^{\text{budget}}$). Additionally, vertical lines are mapped across the grid to mark the exact indifference thresholds between the contract types.

1.4 Output Graphics and Interpretation

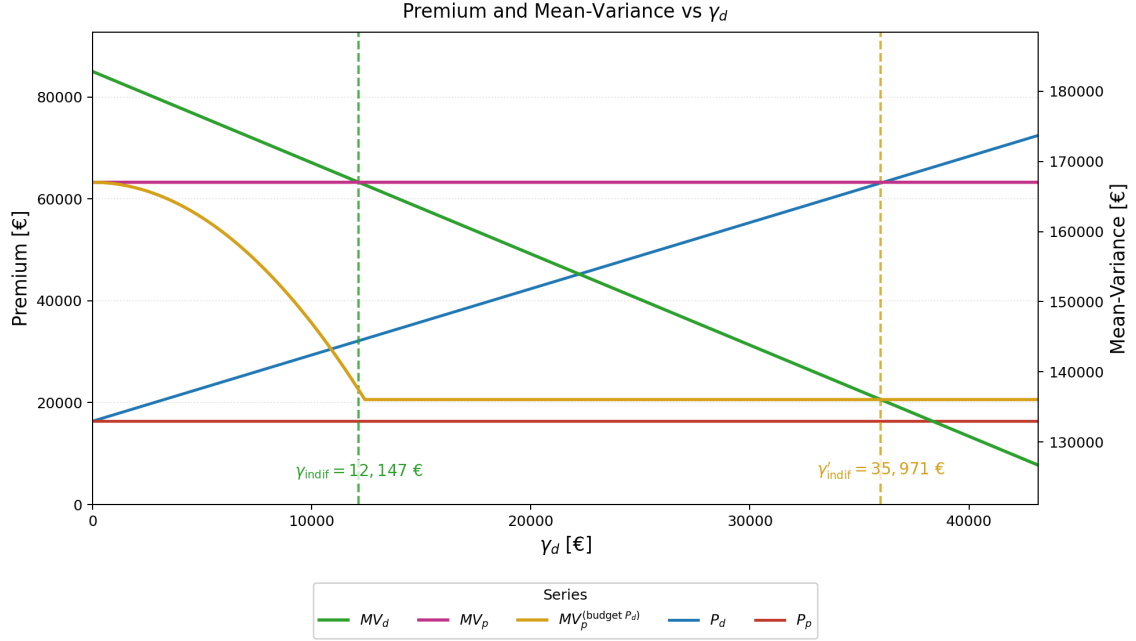


Figure 1: One-dimensional comparative statics in γ_d : premiums and mean-variance at unconstrained optima $d^*(\theta)$ and $k^*(\theta)$. case with $\theta_d = \theta_p = 0.3$.

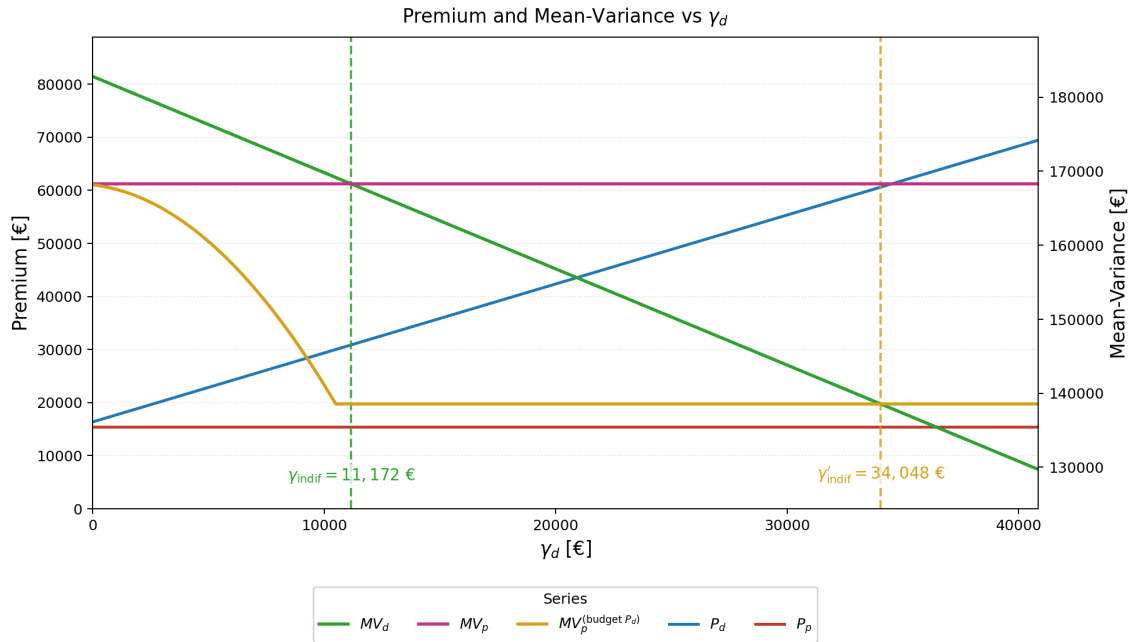


Figure 2: One-dimensional comparative statics in γ_d : premiums and mean-variance at unconstrained optima $d^*(\theta)$ and $k^*(\theta)$. case with $\theta_d = 0.3$ and $\theta_p = 0.2$.

- **Classical Mean-Variance Utility $MV_d(\gamma_d)$:**

The green curve decreases continuously as γ_d rises. Because the classical insurance contract becomes increasingly expensive due to higher indemnity frictions, the utility decreases linearly.

- **Parametric Mean-Variance Utility MV_p :**

The purple line remains constant. Since standalone parametric insurance does not incur any claims verification costs, its risk-utility profile is independent of γ_d .

- **Cost-Equivalent Parametric Utility $MV_p^{(\text{budget } P_d)}$:**

As γ_d increases, the available premium budget expands, causing the implied parametric coverage $k(\gamma_d)$ to continuously grow.

In the first section, the curve decreases quadratically because this growth in $k(\gamma_d)$ causes the underwriting risk to increase, reducing the overall utility.

Beyond the respective threshold, the curve becomes constant, as the coverage amount has reached its absolute upper bound $k(\gamma_d) = L$.

With a lower loading factor ($\theta_p = 0.2$ instead of 0.3), the insured receives more coverage for the same amount of money, which is why this maximum is reached at a lower γ_d and the overall utility level is higher.

- **Classical Premium P_d :**

The blue line increases linearly. This directly reflects the proportionally growing of transaction costs when γ_d increases.

- **Parametric Premium P_p :**

The red line is flat. Because the parametric alternative bypasses the loss-verification process entirely, its premium is decoupled from the indemnity frictions γ_d .

The curves for P_d and $MV_d(\gamma_d)$ remain unchanged in both figures because they are exclusively determined by the parameters θ_d and γ_d , making them independent of changes in θ_p .

The other lines shift numerically when θ_p varies, but their underlying geometric structure remains identical. Specifically, P_p remains a strictly flat horizontal line but shifts downward as the loading factor decreases. MV_p stays horizontal but shifts upward when θ_p decreases.

The budget-constrained utility $MV_p^{(\text{budget } P_d)}$ preserves its characteristic kinked shape.

2 Question 2

2.1 Presentation to Management

The following analysis is intended as a briefing for management of the corporation operating in Cesena. We evaluate and compare the distinct characteristics of classical indemnity contracts and parametric insurance products across seven key dimensions including capital requirements and loading factors, fixed administrative costs, premium levels, payout timing, and reconstruction incentives.

To complement this analysis, we have also filmed a brief elevator pitch. You can access the video recording via the following link: https://drive.google.com/file/d/1LEBJAq_Ly0oKoNV1e90jKnFCtiSgdQsV/view?usp=share_link

2.1.1 Executive Summary: Strategic Recommendation

Our analysis shows that the optimal insurance choice is strongly influenced by the corporation's liquidity position. While indemnity insurance provides full compensation conditional on verified losses, it is associated with significant administrative costs (γ_d) and potentially long payout waiting. These frictions reduce its effectiveness in situations where rapid access to liquidity is critical.

Parametric insurance, in contrast, offers faster and more predictable payouts (e.g., $k^* \approx \text{€}502,244$ within days as obtained in 1.2.2), at the cost of basis risk, since compensation is decoupled from realised losses. In the model, this feature improves post-event liquidity and supports faster recovery decisions, particularly for firms with limited cash reserves relative to asset exposure.

Overall, the analysis suggests that parametric insurance is preferable when the benefits of fast access to liquidity and lower administrative costs outweigh the risk that payouts may not fully match actual losses.

2.2 Lower Capital Requirements and Loading Factors

Traditional insurance policies are subject to what is known as a provision risk. This risk arises from the uncertainty that the actual costs of settling claims may far exceed the amounts estimated and set aside in advance. To absorb potential shortfalls in extreme cases, insurers must maintain substantial capital buffers in accordance with regulatory requirements such as Solvency II. These capital-holding costs are passed on to customers via higher loading factors (θ_d). In the case of parametric insurance, provision risk is structurally absent: payouts are rigidly linked to objective environmental triggers, leaving no actuarial uncertainty post-event. Under the Solvency II standard formula [2], this structurally eliminates the capital charge for the so-called "Reserve Risk" (the risk of adverse development in outstanding claims). As a result, parametric insurers require significantly less regulatory capital, allowing them to offer lower loading factors ($\theta_p \leq \theta_d$).

The quantitative impact of this difference is visible directly in the mean-variance utility. Under symmetric loadings ($\theta_d = \theta_p = 0.30$), the two instruments are on a level playing field with respect to capital cost, and both yield $MV_p = \text{€}167,021.25$ for approach (a). However, when the parametric loading drops to $\theta_p = 0.20$, the parametric utility rises to $MV_p = \text{€}168,288.76$, an improvement of $\approx \text{€}1,268$ reflecting the combined effect of lower pricing distortion and reoptimization for the optimal coverage level k^* . This improvement strengthens the relative attractiveness of the parametric contract and shifts the indifference threshold from $\gamma_{\text{indif}}^{(a)} \approx \text{€}12,147$ to $\approx \text{€}11,172$, meaning the parametric contract becomes optimal at slightly lower levels of indemnity related costs.

It should be noted that a lower θ_p does not come without a counterparty dimension. Parametric contracts are often issued by reinsurers, structures that may carry higher counterparty credit risk than a traditional insurer. The reduction in provision risk is thus partially offset by an increase in default risk of the parametric counterparty, which is not captured in the present mean-variance model. Furthermore, the asymmetric case ($\theta_p < \theta_d$) breaks the structural duality between the two contracts, meaning that cost-equivalence benchmarks derived under symmetry may understate the value of the cheaper parametric product.

2.3 Fixed Administrative Costs

By eliminating the administrative costs ($\gamma_p = 0$), a parametric framework delivers two decisive economic advantages.

First, it provides a structural premium advantage. Because traditional insurers must absorb heavy claim and verification costs (γ_d), these inflate premiums at rate $(1 + \theta_d)\gamma_d$. The direct effect on the insured company's mean-variance utility is:

$$MV_d(\gamma_d) = 182,812.97 - 1.3\gamma_d$$

so that every $\text{€}1,000$ of auditing cost reduce $\text{€}1,300$ of mean-variance utility. In the context of a 1 M asset with flood frequency $\lambda = 1/40$, the approach (a) threshold of $\approx \text{€}12,147$ is easily exceeded by a industrial claim involving specialist engineering surveys or legal disputes.

Secondly, it provides the insured company with a competitive advantage in terms of human capital by avoiding opportunity costs. Under a classical indemnity framework, specialised engineers, corporate attorneys, and financial officers are heavily tied up for months compiling historical documentation and proving loss causality to sceptical claims adjusters. Under a parametric contract, payouts trigger automatically via independent data feeds, freeing up key corporate resources to focus on operational recovery.

The assumption $\gamma_p = 0$ holds in the model but is an idealisation. In practice, designing and maintaining a parametric contract requires upfront legal structuring, ongoing data-provider fees, and periodic trigger recalibration. It should be verified that the contractual and data-infrastructure costs of the parametric product are genuinely negligible relative to the expected γ_d savings before concluding that $\gamma_p \approx 0$ is a safe assumption for their specific facility.

2.4 Low Premium Insurance

As explained in the two points above, the lower capital requirements under Solvency II and the elimination of ex-post regulatory costs ($\gamma_p = 0$) result in lower premiums for parametric insurance. A factor

that further reinforces this lower premium is the elimination of information asymmetries, in particular moral hazard (behavioural risk).

In traditional insurance, the insurer needs to take into account safety margins, because of the risk that the customer will act less carefully after the contract is concluded. In the parametric context, this problem is completely eliminated. The payout is rigidly linked to an objective data trigger and is paid out regardless of the actual loss, the company retains the maximum incentive to protect itself against losses.

From the perspective of a risk-averse company, this cost efficiency cannot be evaluated in isolation. It must be weighed against the structural disadvantage of basis risk. While a classical contract guarantees full financial indemnity (above the deductible) and eliminates volatility regarding the actual loss, the parametric contract decouples the payout from the true physical damage. This introduces the critical threat of a mismatch. The corporation may suffer a catastrophic regional event, but receive zero compensation if the specific environmental trigger threshold is missed.

Therefore, the management faces a fundamental economic trade-off. Choosing parametric insurance means trading a deterministic, costly administrative friction (γ_d) for a stochastic, potentially volatile residual risk. In a pure risk-neutral budget framework, the savings from eliminating indemnity frictions can easily offset the average expected cost of unhedged losses. For a highly risk-averse management, the fear of severe, uncompensated tail-events via basis risk penalizes the utility of the parametric product. Consequently, parametric insurance only becomes the optimal corporate choice if the internalized administrative savings and immediate liquidity velocity are large enough to overcompensate for the subjective disutility of the underlying basis risk.

To make an strategic decision, one should consider the indifference points, as illustrated in the plots above. The indifference points can be viewed from two different perspectives: total utility efficiency in green and pure budgetary burden in yellow. For the company's final decision, the indifference point of total utility should primarily be used, because this indifference point defines the true economic welfare frontier, where the two mean-variance curves intersect. If the traditional claims settlement costs ($\gamma_d \approx 12\,147$ or $\gamma_d \approx 11\,172$) exceed this threshold, the parametric product fully offsets the inherent basis risk through its superior lack of friction. And therefore offers the mathematically and operationally basis for decision-making for a risk-averse management seeking to maximize its total utility.

2.5 Timing of Payouts

An important divergence between classical indemnity and parametric insurance lies in payout speed and its impact on corporate liquidity. Traditional indemnity contracts follow a multi-stage process: formal loss notification, adjuster dispatch, survey completion, and final loss-quantum agreement. For complex industrial claims this process regularly takes months. During a regional catastrophe such as a major flood across the Emilia-Romagna plain, simultaneous filings strain insurer capacity, further compounding these delays.

Parametric insurance triggers mechanically the moment a pre-agreed threshold is crossed, for example, when the Savio river at Cesena exceeds a specific gauge reading for a minimum duration, and payouts are typically disbursed within days of the event being confirmed by the independent data provider.

The liquidity dimension is particularly acute for our model corporation. With initial wealth $\omega_0 = \text{€}200,000$, even a moderate flood loss of $\text{€}300,000$ would wipe out the corporation's entire liquid reserves before any classical insurance settlement arrives. A parametric payout of $k^* = \text{€}502,244$ received within days of the event would instead cover the loss in full and leave the corporation with a positive cash balance, allowing it to secure emergency contractors and maintain supply-chain continuity.

2.6 Real World Scenario Analysis: The 2023 Emilia Romagna Flood

To move beyond abstract theoretical indifference points, we contextualize our findings within the documented reality of the May 2023 flooding events in the Emilia Romagna region (e.g., the Savio river overflow). According to the official estimates certified by the European Union and the Regional Government, the disaster caused approximately $\text{€}8.5$ billion in total damages, severely disrupting the local productive sector.

The mathematical framework of our problem assigns the corporation a total value at risk of $L =$

€1,000,000 (representing an industrial shed with specialized machinery) and a highly constrained liquid initial wealth of $\omega_0 = €200,000$.

The Classical Indemnity Reality (Post Event Friction):

Following the 2023 event, both traditional insurance companies and the government compensation structure (governed by the *Ordinanza n. 11/2023 del Commissario Straordinario*) required a sworn expert appraisal (*perizia asseverata*) to strictly certify the causal link and the exact quantum of damages before releasing full funds. According to post-flood reports by *Confindustria Romagna*, this rigorous loss-adjustment process forced companies to anticipate massive reconstruction costs. For our model corporation, facing a severe flood, the classical contract's theoretical "full coverage" above the $d^* = €30,000$ deductible becomes a fatal liquidity trap. The company risks insolvency during the months required to complete the auditing process (γ_d), because its €200,000 cash reserve (ω_0) is insufficient to resume operations and retain market share while waiting for the payout.

The Parametric Solution (Liquidity Velocity):

Under the parametric contract ($k^* \approx €502,244$), the payout is entirely decoupled from the *perizia asseverata*. If the river gauge breaches the predefined threshold, the data provider verifies the event within days. The immediate injection of half a million euros directly overrides the ω_0 liquidity constraint. Management can immediately secure emergency contractors and replace damaged machinery, entirely bypassing the administrative friction (γ_d) that paralyzed the region's classical recovery efforts.

Strategic Conclusion: For a low-liquidity corporation ($\omega_0 \ll L$), the speed of parametric liquidity is a critical survival mechanism. The administrative cost γ_d in classical contracts is not just a monetary fee, but a proxy for operational paralysis.

2.7 Reconstruction Incentives

A fundamental divergence between classical indemnity and parametric insurance lies in the post-loss reconstruction incentives. In a classical indemnity framework, the contract is strictly governed by the indemnity principle, which prohibits the insured from financially profiting from a loss. Consequently, the insured corporation does not receive a payout to spend at its discretion but must prove actual expenditures to the insurer's adjuster. Once a loss benchmark is established, the corporation has zero economic incentive to minimise reconstruction costs below that benchmark, since any savings would flow back to the insurer. Aware of this misalignment, classical insurers are forced to invest heavily in post-loss monitoring, auditing and verification, generating the γ_d cost channel analysed throughout this project.

Parametric insurance bypasses this costly verification cycle. Because parametric payouts are decoupled from individualised loss assessments and trigger automatically based on objective environmental events, the insured receives a fixed payment. This structural freedom provides optimal reconstruction incentives. If the payout is deployed strategically, negotiating competitive contractor rates or adopting flood-resilient building technologies such as elevated flooring, sump pumps, or flood barriers, any saving below k accrues entirely to the corporation, rewarding operational agility.

The reconstruction-incentive advantage of parametric insurance collapses precisely when it is needed most. Under the censored exponential severity distribution with $\nu = 1/700,000$ and $L = €1,000,000$, the probability of censoring at the maximum loss is:

$$\Pr(Y_i = L) = e^{-\nu L} = e^{-1/0.7} \approx 23.6\%$$

In any such event, the corporation receives only $k^* = €502,244$ but faces reconstruction costs of up to €1M. The shortfall of up to €497,756 must be funded from initial wealth ω_0 , leaving a capital gap of up to $\approx €297,756$ that the corporation cannot bridge without external financing. In this scenario the reconstruction incentive becomes irrelevant. The corporation may be forced to partially abandon the facility rather than rebuild it. This is the ultimate cost of basis risk, and the primary reason the mean-variance criterion still assigns positive value to the classical contract at low γ_d levels, despite all the friction-based disadvantages described above.

2.8 Extra: Insuring the Uninsurable

A major advantage of parametric insurance is its ability to cover indirect business interruptions that are virtually uninsurable under traditional frameworks, as demonstrated by regional infrastructure risks in Emilia-Romagna. Consider a severe flooding event in the Cesena region, where rising water levels breach a local utility station, causing a widespread power grid collapse. A manufacturing corporation in Cesena may suffer devastating financial losses due to the resulting production halt, even if its own facilities remain completely dry and unscathed by the floodwaters. Under a classical indemnity contract, the firm would receive zero compensation because there is no direct physical damage to its insured assets. A parametric contract, by contrast, triggers automatically based on the regional event such as a verified river gauge level or a documented grid outage duration. In this way business interruptions that arise when regional infrastructure fails, get mitigated through liquidity.

2.9 Extra: Model Robustness and Fat Tails (Extreme Value Theory)

As Professor Gardini discussed regarding stochastic models (Variance Gamma and Jump Processes), standard models relying on exponential dynamics tend to severely underestimate the frequency of extreme events. Real-world catastrophic risks, such as major floods, exhibit *Fat Tails*. To test the robustness of our Mean-Variance framework against this model risk, we substitute the censored Exponential distribution with a censored Pareto (Lomax) distribution, which is the standard in Extreme Value Theory (EVT).

The survival function of the Pareto distribution is $S(y) = \frac{\theta}{\theta+y}$.

To ensure a rigorous and fair comparison, we strictly calibrate the scale parameter $\theta \approx 462,047.52$ (using the numerical solver `scipy.optimize.fsolve`) so that the expected damage remains *identically equal* to our baseline scenario:

$$\mathbb{E}[Y_{Pareto}] = \theta \ln \left(1 + \frac{L}{\theta} \right) = \text{€ } 532\,244.14 \quad (31)$$

Theoretical invariance: By setting the first-order conditions for the Pareto distribution, we discover a remarkable theoretical property: the optimal contract parameters k^* and d^* depend entirely on the first moment $\mathbb{E}[Y]$ and the loading factors, making them strictly invariant to the tail shape of the severity distribution.

Economic impact of Basis Risk: While the optimal thresholds do not change, the second moment $\mathbb{E}[Y^2]$ explodes due to the heavy tails. Because the parametric payout is capped at k^* , the increased probability of extreme loss L drastically amplifies the residual Basis Risk. As visualized in our supplementary Python *Fat-Tail Stress Test* (see Figure 3), this explosion in variance structurally destroys the utility of the parametric insurance. Under Fat Tails, the corporation will rationally prefer the classical indemnity contract even when confronted with high indemnity frictions (γ_d), highlighting a critical vulnerability of parametric triggers in extreme climate scenarios.

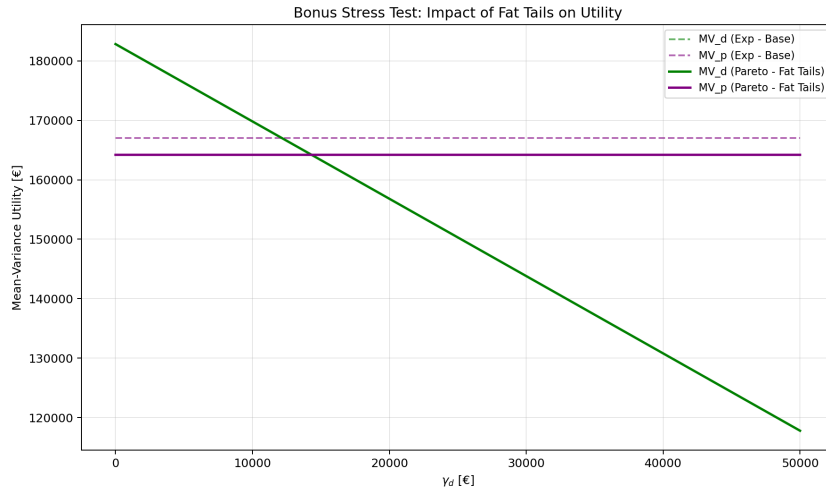


Figure 3: Stress Test: Premiums and Mean-Variance under a Fat-Tailed (Pareto) severity distribution with identical expected loss. The Basis Risk drastically penalizes parametric utility.

References

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